The Anchor Accumulation Problem: Bounded Anchor Sets, a Soft–Hard Protection Law, and the Auditable Cost of Regularized Continual Learning

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Abstract

Regularized continual-learning schemes that store one protection state per task pay a cost that grows with sequence length. In usage-localized elastic weight consolidation (DeLP), per-neuron elastic anchors persist at full weight as an elastic drag on later learning; a retention decomposition $\bar{a}_T = \ell + \frac{T-1}{T}\,\mathrm{BWT}$ localizes the cost in the *learning term* ℓ. We call this the *anchor accumulation problem* and study its repair: bounding the anchor stack at M entries. A FIFO budget recovers +4.5 and +9.0 points of final accuracy over unbounded DeLP at T=10 and T=20 (M=5; n=10 paired seeds; Wilcoxon p=0.002), and the budget is a distinct knob from the elastic weight λ: it controls the *age structure* of forgetting, λ its *amount*. Eviction design is ruled by one constraint: any criterion keyed to current constraint mass grandfather-protects stale anchors and evicts new ones at birth (−34.5 BWT), whereas age-first criteria are immune. Sweeping online EWC's discount shows that discounting is itself a *soft* bound, and soft and hard bounds obey one log-linear law between the learning term and the protected-mass horizon ($\ell \approx$

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$84 - 11 \ln h$, R^2=0.99): bounding accumulated anchor mass, not eviction, is the active ingredient, and the explicit window retains what a scalar discount cannot — a discrete, named, policy- selectable protection set readable from the stack trace. We report the boundary honestly: on Split-CIFAR-100 under task-incremental evaluation, a no-protection baseline stays above every protection arm, so the budget result is internal to the protected family; the paper's contribution is a *knowledge-based diagnosis and audit framework* — when to protect, how to bound, which anchors survive — not a drop-in accuracy win.

Keywords: continual learning, elastic weight consolidation, catastrophic forgetting, knowledge-based protection, stability–plasticity dilemma

1. Introduction

Elastic weight consolidation (EWC) [1] and its variants protect important parameters by penalizing departure from values that served previous tasks. The idea originates in the computational principles of synaptic memory consolidation [2], and a long line of work has since shaped it: synaptic intelligence [3] accumulates path integrals online, memory-aware synapses [4] estimate sensitivity without supervision, and context-dependent gating [5] makes the protection pathway itself task-conditional. Surveys of the field [6, 7] document both the breadth of these designs and how few of them are comparable under a common protocol — a gap our own audit of DeLP [8] ran into directly.

From a knowledge-based-systems perspective, the protection state that these methods maintain *is* a knowledge base: a growing, structured store of

what has been learned, consulted at every update, and trusted in proportion to its age. Yet this knowledge base is rarely treated as a first-class design object — it is appended to, never curated, bounded, or audited. This paper makes it the object of study.

DeLP [8] localizes this protection to dendritic compartments and weights each neuron's penalty by its measured usage under previous tasks. The audit of the method established that usage weighting survives a parameter-budget cap (+1.79, p=0.005 at twenty tasks) while the full dendritic chain adds little once baselines share the hidden activation.

The same audit exposed a structural cost: DeLP appends one anchor set (posterior means and Fisher estimates) per task, so the elastic drag on optimization grows with the sequence length T. A decomposition of average retention,

$$\bar{a}_T \;=\; \ell \;+\; \frac{T-1}{T}\,\mathrm{BWT}, \tag{1}$$

separates the *learning term* ℓ (how well each task is acquired at its boundary) from the backward-transfer term. At ten tasks the losses of unbounded DeLP are concentrated in ℓ: anchors from early tasks persist as elastic drag long after the tasks they protect have been consolidated, and this is a property of the protection scheme, not of the architecture.

This paper treats the anchor stack itself as the object of study. We bound it at M entries and ask four questions: **(Q1)** Can the learning term be recovered by bounding, and at what stability cost? **(Q2)** Is the budget a redundant knob (a non-linear re-scaling of λ), or does it control something λ cannot? **(Q3)** If the stack must be bounded, *which* anchors should be evicted? **(Q4)** Does protection itself pay for its cost — or does a no-

protection reference beat the entire protected family?

Contributions.. Our contribution is a *diagnosis-and-audit framework* for protection state in regularized continual learning, together with a quantitative account of when protection is worth its cost. We state the boundary of the repair up front as a finding, not a footnote:

1. **A protection-must-justify-itself result.** On Split-CIFAR-100 under task-incremental evaluation, a no-protection baseline (λ=0) stays above every DeLP arm including the best budgeted one (Sec. 5.9). The decomposition explains why: at any λ, protection roughly halves the learning term while retention in this regime is nearly free to forgo. We therefore credit no protection benefit on that benchmark and recommend a no-protection reference as a mandatory check for any regularized CL pipeline — a negative result we argue increases, rather than decreases, the audit value of the framework.
2. **The anchor accumulation problem and its bounded repair.** We identify the accumulation of full-weight per-task anchor stacks as the binding constraint on the learning term of long sequences, and introduce a family of eviction policies for anchor stacks. A FIFO budget recovers $+4.5/+9.0$ points of final accuracy at T=10/20 over unbounded DeLP (n=10 paired seeds; Wilcoxon p=0.002), by recovering $+5.4/+15.2$ points of the learning term (Eq. 1); the effect replicates on Fashion-MNIST (+13.0 at M=2) and, within the protected family, on Split-CIFAR-100 (five of six bounded policies beat unbounded at T=20, M=2, up to +5.0 points, Wilcoxon p=0.002).
3. **An eviction design space with one governing constraint.** Through

per-task accuracy profiles we show the budget controls the *age structure* of forgetting while λ controls its total amount (Fig. 2); and trace instrumentation (Figs. 4, 5) shows that any criterion keyed to an anchor's *current constraint mass* fails systematically: it grandfather-protects stale anchors and evicts new ones at birth (BWT -34.5). Multiplicative age re-weighting does not fix this; an age-first, mass-second criterion does. The primary key, not the secondary signal, determines whether a criterion works.

4. **A soft–hard protection law.** Sweeping online EWC's discount γ shows a tuned discount overtakes the budget on accuracy at matched forgetting — and that discounting is itself a *soft* bound. Across soft (discount) and hard (window) bounds, the learning term falls log-linearly in the protected-mass horizon ($\ell \approx 84 - 11 \ln h$, R^2=0.99, per-seed $R^2 \geq 0.98$, monotone at every grid λ; the same four-arm fit at T=10 gives a shallower slope, -3.8, R^2=0.96): bounding accumulated anchor mass, not the mechanism enforcing it, is the active ingredient.
5. **Auditability as the residual value of explicit windows.** Because the budget is a *set* of named anchors rather than a scalar discount, the protected mass is inspectable: which tasks are still protected, which were evicted, and by which policy, are all readable directly from the stack trace, and the forgetting profile the trace predicts is the one measured (8.6 mean drop on the five in-window tasks vs. 27.7 on the five oldest evicted tasks). Fisher-cosine merging further decouples memory from behavior: the penalty *value* drops one-sidedly (up to 64%, gradient unchanged) while accuracy stays indistinguishable from un-

bounded. What the explicit window retains over a tuned discount is thus exactly the knowledge representation: a discrete, named, policy-selectable protection set.

2. Related work

Regularization-based continual learning.. EWC [1] places a quadratic penalty around posterior estimates of previous tasks; SI [3] accumulates path-integral importances online; MAS [4] estimates sensitivity unsupervised. Anchor accumulation is not specific to DeLP: any scheme that stores per-task quadratics pays storage $O(T)$ and, if all terms persist at full weight, an increasingly rigid objective. Online EWC [9] addresses the storage axis by collapsing the stack into a single running posterior; with a discount $\gamma<1$ its Fisher mass is geometrically bounded, which makes it a *soft* bound on the same accumulated mass we diagnose (Sec. 5.2). Bounded anchor sets are complementary: they keep per-task terms and bound their number.

Recent continual-learning methods.. Recent work treats the stability–plasticity trade-off from several angles, and our work is complementary to all of it because it targets the *accumulation mechanism* of a regularizer rather than the network parameterisation. Forget to Learn [10] circumvents the conflict in continuous unsupervised domain adaptation by active forgetting plus conflict-free distillation; TIPS [11] selects prompts at two granularities to rebalance stability and plasticity in prompt-based continual learning; a dual-learner with cumulative parameter averaging [12] pairs a plastic learner with a stable one; and LoRAC [13] constrains critical parameter changes under orthogonal LoRA tuning. These are all *architectural or replay-side*

remedies. Ours is complementary: we ask what happens *inside* a regularizer that accumulates one anchor set per task, and we show that the accumulation itself — not the eviction rule, not the discount schedule — is what costs plasticity. We cite the rehearsal-side family [14] as the reference point that makes exemplar-free regularizer comparisons legible: a strong baseline of this kind forces one to state honestly in which regime a comparison holds.

Parameter isolation and budgeting.. PackNet [15] and HAT [16] budget capacity explicitly. Our contribution differs: the budgeted object is the *protection* state, not the network capacity, and eviction can be nearly free when the penalty is combined correctly (merge).

Dendritic computation.. The dendritic substrate is not decorative. Pyramidal neurons are multicompartmental systems in which local dendritic potentials shape integration at the soma [17, 18, 19], and these computations are measurable *in vivo* rather than merely modeled: dendritic activity in the cortex is multiplexed and globally modulated [20], and it computes abstract rules [21]. Reviews of dendritic computation [22, 23] and of its engineering exploitation [24, 25, 26, 27] make the relevant point for us: the compartment is a computational unit with its own timescales, so protecting it with a task-wise snapshot is a substantive modeling choice rather than an implementation detail. Active-dendrite networks [28, 29] provide the architectural substrate; DeLP's audit [8] established which part of the dendritic chain carries the benefit.

Scope of this manuscript in relation to the author's precursor work.. Two earlier Zenodo preprints by the same author [8, 30] audited, respectively,

which component of the dendritic chain carries the DeLP benefit and how that benefit distributes across input segments. Both are diagnostic studies of *where* the benefit originates; neither examines what happens to a stack of anchors as the task sequence grows. The present manuscript takes the opposite direction. Its subject is the cost that *any* stack-building scheme pays over a long sequence — the anchor accumulation problem — and its object is the repair for that cost: bounding the anchor stack at M entries, the eviction criteria that decide which anchors survive, and the resulting law relating soft and hard protection bounds. All experiments, analyses, figures, and text in the present manuscript are new; the precursor audits are cited as prior work and are not extended, reproduced, or reused here.

3. Background: DeLP and anchor accumulation

Setup.. A network is trained on a sequence of T tasks (permuted MNIST / Fashion-MNIST). DeLP freezes dendritic segments unused by the current task and adds, after each task t, an anchor set $\mathcal{A}_t = \{(\theta_t, F_t)\}$ (parameter snapshot and Fisher diagonal). The protection objective during task t is

$$\mathcal{L}_t(\theta) \;=\; \mathcal{L}_t^{\text{task}}(\theta) \;+\; \frac{\lambda}{2} \sum_{k<t} \sum_{i} F_i^{(k)} \, (\theta_i - \theta_i^{(k)})^2, \tag{2}$$

where the inner sum runs over the accumulated stack $\mathcal{A}_1, \ldots, \mathcal{A}_{t-1}$: the penalty is *unbounded* in T, both in storage and in constraint mass.

Where the cost lands.. Following [8], average retention decomposes as Eq. 1. At fixed λ=5000 the learning term of unbounded DeLP falls from 59.2 at T=10 to 37.7 at T=20 — a 21.5-point drop that compounds with sequence length, exactly the signature of drag that accumulates with every finished

task; anchored identity Eq. 1 attributes the gap to ℓ, not BWT. We call this the *anchor accumulation problem*.

4. Bounded anchor sets

Budgeted stacks.. We replace the unbounded stack with a bounded one of size M. After appending the anchor for task t, if $|\mathcal{A}| > M$ an *eviction policy* removes or combines entries:

$$\mathcal{A} \leftarrow \mathsf{evict}_{\pi}(\mathcal{A} \cup \{(\theta_t, F_t)\},\, M)\,. \tag{3}$$

Policies.. Table 1 defines the six policies we compare. They span the design space along one axis: *what signal drives the choice* — age (fifo), chance (random), current constraint mass (redun), pairwise redundancy (merge), or combinations of age and mass (agew, agew2).

Instrumentation.. Each anchor carries an *origin* tag (the task that created it), and we record the surviving-origin set after every phase (`survivor_history`). This turns the eviction dynamics into a directly observable object (Fig. 4).

5. Experiments

Protocol.. Permuted-MNIST and Fashion-MNIST, $T \in \{10, 20\}$; architecture, optimizer, permutation generation, and evaluation identical to the DeLP audit protocol [8]; λ=5000 unless stated. Every arm runs at the pipeline's *default* hidden-layer gain — the uncalibrated condition of the audit's calibration protocol [8] — and no arm is gain-tuned within this study; since the gain is shared, the confound the audit documented (a single gain

Table 1: Eviction policies for a stack over budget. Constraint mass $m_k = \sum_i F_i^{(k)}(\theta_i^{\text{now}} - \theta_i^{(k)})^2$ is measured at eviction time.

Policy	Key	Rule (when $\lvert\mathcal{A}\rvert > M$)
unbounded	—	never evict (original DeLP)
fifo	age	evict the oldest anchor
random	—	evict uniformly at random
redun	mass	evict $\arg\min_k m_k$
agew	mass × age	evict $\arg\min_k m_k \cdot (k{+}1)$, oldest $k{=}0$
agew2	age → mass	evict oldest anchor with below-median m_k; fallback fifo
merge	redundancy	merge the pair with most similar F (cosine): $F \leftarrow F_i + F_j$, $\theta \leftarrow$ weighted mean; repeat

scalar moves accuracy by more than the entire method) cannot differentially favor any arm, only shift all arms together. Paired comparisons use identical seeds. Budgets $M \in \{2, 5\}$ anchor the main grid; a seven-point dose–response sweep $M \in \{2, 3, 4, 5, 6, 8, 10\}$ (T=20, λ=5000; Sec. 5.4) fills the interior, and no M was tuned post hoc. Section 5.8 states the Split-CIFAR-100 protocol separately (different backbone and protection pathway).

5.1. Bounding recovers plasticity (Q1)

Table 2: a FIFO budget at M=5 trades 1.5/7.2 points of BWT for +4.5/+9.0 accuracy. The mechanism is the learning term: ℓ rises by +5.4 and +15.2, exactly the recovery predicted if accumulated elastic drag were the binding constraint (Sec. 3); the grid in Table 3 shows the same signature for every bounded arm. Eq. 1 is an accounting identity, so the decomposition alone is

Table 2: Paired fifo–unbounded deltas, M=5, λ=5000, n=10 seeds. $t(9)$ is the paired t statistic; W_p is the two-sided p-value of the Wilcoxon signed-rank test on the same paired deltas, and W_p^{SW} is the Shapiro–Wilk p-value on those deltas (normality of the paired difference, on which the t test depends).

Arm	Δacc	$t(9)$	W_p	W_p^{SW}	ΔBWT
fifo T=10	$+4.51 \pm 0.67$	21.4	0.002	0.024	-1.50 ± 0.49
fifo T=20	$+8.98 \pm 1.49$	19.1	0.002	0.992	-7.20 ± 1.97

not evidence; the causal claim rests on the eviction traces (Sec. 5.5) and on the λ-by-budget interaction (Sec. 5.4), which localize *when* the learning term moves — namely when anchors are actually retired. Corrected SI at T=10 (λ=100) reaches 40.3%.

5.2. Single-anchor schemes: the stack vs. its alternatives

Because the diagnosis targets the *stack*, we also ran two protection schemes that cannot accumulate one. Both use the identical architecture, protocol, and λ grid (n=10 paired seeds): plain EWC, which keeps only the most recent task's Fisher and anchor (no accumulation, no running Fisher update — distinct from the original accumulated variant of [1]), and online EWC [9] with $\gamma \in \{0.5, 0.9, 1\}$, which collapses the stack into a single running posterior with geometrically discounted Fisher accumulation. All arms share the identical λ grid and are compared at each scheme's best grid value, so no scheme is disadvantaged by another's tuning.

Plain EWC posts the highest final accuracy of any arm (68.3–71.1 at T=10, 56.4–59.4 at T=20, best at λ=1000) by maximizing ℓ (81–85): with no stack there is no accumulation — and no retention either, at 3–5× the

Table 3: Policy grid, M=5, λ=5000, mean ± sd over n=3 seeds. ℓ = learning term (mean accuracy at task boundaries). agew is bit-identical to redun, as predicted in Sec. 5.5.

T	Arm	BWT	acc	ℓ
10	unbounded	-5.5 ± 0.6	55.3	60.2
10	fifo	-6.6 ± 0.9	59.7	65.6
10	random	-9.1 ± 2.6	60.2	68.4
10	redun	-18.3 ± 0.8	53.6	70.1
10	agew	-18.3 ± 0.8	53.6	70.1
10	agew2	-8.4 ± 0.9	60.8	68.3
10	merge	-5.5 ± 0.6	55.3	60.3
20	unbounded	-4.4 ± 1.0	33.7	37.9
20	fifo	-10.2 ± 1.9	43.4	53.1
20	random	-13.5 ± 0.4	42.0	54.8
20	redun	-30.2 ± 0.4	36.3	65.0
20	agew	-30.2 ± 0.4	36.3	65.0
20	agew2	-16.5 ± 0.3	42.3	58.0
20	merge	-4.5 ± 0.9	33.6	37.9

forgetting of unbounded DeLP (BWT $-15.8/-25.7$ at its best λ vs. $-5.5/-4.4$). Under the final-accuracy metric the learning-term dominance outweighs retention; a reader who weights BWT equally would rank it last.

Online EWC at its best grid value (λ=1000) reaches 67.5/46.0: at T=10 it is statistically indistinguishable from DeLP-fifo at the same λ (67.4; paired $t(9)$=-0.3); at T=20 fifo is +3.5 ahead ($t(9)$=7.0). A collapsed posterior thus recovers most of the plasticity, but it posts neither the best long-sequence

accuracy nor, by construction, any anchor accumulation problem to cure. Its forgetting is not unstructured: per-task forgetting (peak minus final accuracy; T=20, λ=1000, n=10) concentrates on older tasks for every arm, with drop–age correlations of -0.94 (plain), -0.79 (DeLP-fifo), -0.70 (online EWC), and -0.43 (unbounded).

5.3. The soft–hard protection law

Sweeping the discount shows that the collapsed posterior does have an age knob after all: lowering γ to 0.9 and 0.5 raises ℓ from 51.1 to 56.6 and 76.1 (T=20, λ=1000), and at γ=0.5 online EWC overtakes DeLP-fifo at every grid cell we tested (paired +4.7 to +7.9, all $|t(9)| \geq 10.4$) at essentially matched BWT (-19.5 vs. -19.7). Discounting, however, is itself a bound: with geometrically decayed Fishers the accumulated mass converges to an effective horizon of $\approx 1/(1-\gamma)$ tasks, so $\gamma<1$ is a *soft* counterpart of the eviction window.

The two mechanisms turn out to be interchangeable points on one curve. Across all five bounded-mass arms at λ=1000, ℓ is log-linear in the protected horizon h (h=1 plain, h=2 for γ=0.5, h=5 for the window, h=10 for γ=0.9, h=20 unbounded):

$$\ell \;\approx\; 84 - 11 \ln h, \qquad R^2 = 0.99, \tag{4}$$

a fit that holds for every seed individually (per-seed $R^2 \geq 0.98$, slope -10.9 ± 0.6); the same four-arm fit at T=10 (excluding γ=0.9, whose ℓ sits 0.7 below the line) gives a shallower slope (-3.7, R^2=0.95), consistent with a 10-task sequence probing a shorter stretch of the horizon; and the ordering is monotone at every grid λ (mean-fit R^2 0.99/0.94/0.80 as λ tightens). Bounding

accumulated anchor mass — not the mechanism that enforces the bound — is the active ingredient, and the soft bound is the stronger remedy on final accuracy in this regime.

What the explicit window retains is auditability and policy-level choice: its protection set is discrete and named by the survivor trace; the forgetting profile it produces is the one the trace predicts (8.6 mean drop on the five in-window tasks vs. 27.7 on the five oldest evicted tasks; T=20, λ=1000); and the eviction key can be swapped (age, mass) or the stack merged without touching the update rule. The object studied here remains the stack; the budget is one of two cures, and on this benchmark the simpler one wins.

5.4. Budget vs. λ: two orthogonal knobs (Q2)

Fig. 1 shows the Pareto plane; Fig. 2 the per-task accuracy profiles; Table 4 the λ interaction.

Dose–response.. A seven-point sweep at T=20, λ=5000 (n=10) shows the budget's effect is monotone across the whole interior, not an artifact of two extreme points: final accuracy $52.1, 48.9, 45.0, 42.4, 40.2, 36.5, 35.1$ for M=$2, 3, 4, 5, 6, 8, 10$, with BWT recovering $-25.0 \rightarrow -5.2$ as the window widens toward M=10; every M remains significantly above unbounded at the same λ=5000 (33.4; paired $t \geq 9.1$ at every M, from t=55.4 at M=3 down to t=9.1 at M=10, approaching the window's own limit).

The fifo advantage is robust to the choice of λ: even granting unbounded DeLP its best grid value at each T (λ=1000, the value at which unbounded is strongest at T=20), fifo retains a +10.9-point advantage there (and +3.5 at T=10). That the unbounded-optimal λ must drift downward as T grows

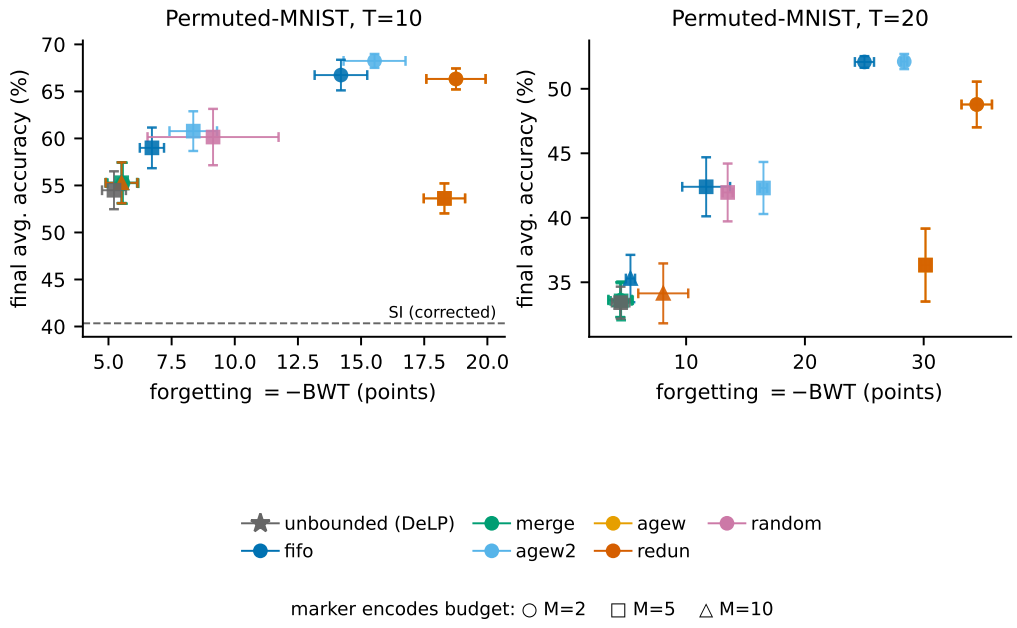


Figure 1: Accuracy–forgetting trade-off on Permuted-MNIST. Color encodes policy; marker encodes budget M. Error bars: sd over seeds (n=10 for unbounded/fifo M=5; else n=3). The smaller budget M=2 trades BWT for accuracy at both budgets in the same direction; merge coincides with unbounded at every M; redun sits dominated.

is itself a symptom of accumulating rigidity: the protection budget has to be relaxed sequence-wide to compensate for anchors that a bounded scheme would simply retire.

Both knobs move the system along a protection–plasticity spectrum, but they are not re-parameterizations of one another. At matched plasticity recovery, the two knobs produce *different age structures* of forgetting (Fig. 2): λ=1000 scales down protection uniformly — the profile keeps its steep age gradient (41.8-point spread, oldest tasks at 66–83%, newest at 41–50%) — while fifo M=2 redistributes forgetting onto the oldest tasks, flattening the

Table 4: λ interaction (M=5, n=10 paired seeds): the fifo advantage is positive at every λ and grows with T (all paired $t(9) \geq 11$, $p < 10^{-6}$).

λ	T	unbounded BWT/acc	fifo BWT/acc	Δacc
1000	10	−11.6/63.9	−10.8/67.4	+3.5
5000	10	−5.2/54.5	−6.7/59.0	+4.5
20000	10	−1.9/49.0	−4.0/53.1	+4.1
1000	20	−13.8/38.6	−19.7/49.5	+10.9
5000	20	−4.5/33.4	−11.7/42.4	+9.0
20000	20	−1.5/29.8	−6.6/37.3	+7.5

profile (22.1-point spread). No value of λ produces the flat profile, and no value of M produces the uniform scaling.

5.5. Which anchors to evict (Q3)

The grandfathering trap.. Mass-keyed eviction is not a straw man: "drop the anchor that constrains least" is the natural heuristic, homologous to low-sensitivity-first intuitions in pruning and in MAS-style importance estimates, and it is what a practitioner would try first. Its systematic failure is the point. Constraint mass m_k (Table 1) is measured at eviction time and is a function of staleness: anchors created before the budget first binds have $\theta^{(k)}$ far from θ^{now} and permanently large m_k; anchors created *after* the stack is full have $\theta^{(k)} \approx \theta^{\text{now}}$ and $m_k \approx 0$. Mass-keyed eviction therefore preserves exactly the stalest anchors and evicts new ones *at birth*: redun's stack freezes at $\{\mathcal{A}_1, \mathcal{A}_2\}$ (Fig. 4), the first two tasks keep strong protection (87/85% final accuracy at M=2), and tasks created later lose all protection within one phase (final accuracy 19–25%), driving BWT to −34.5 at T=20 despite the *highest* ℓ of

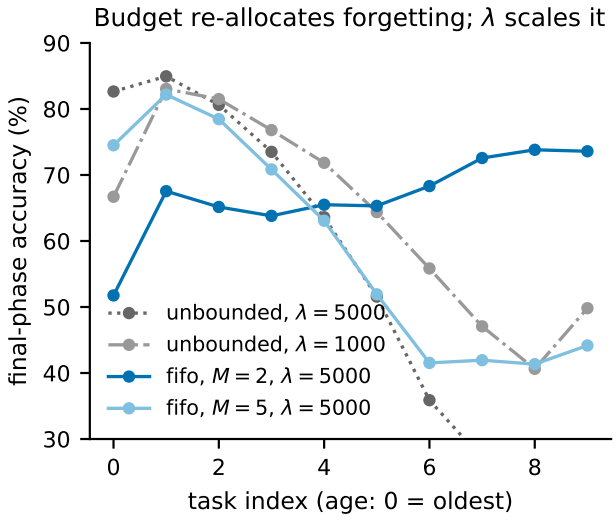


Figure 2: Final-phase accuracy per task (T=10, n=10). λ knobs (gray) translate the profile; budget knobs (blue) flatten it: forgetting is re-allocated to the oldest tasks.

any arm.

Re-weighting mass by age does not help.. agew (mass $\times$ age-rank) is *bit-identical* to redun across all seeds and configurations: mass differences span orders of magnitude and swamp the age factor, so the argmin never flips. Any criterion with mass as the primary key inherits the trap.

Age-first criteria close the matrix.. agew2 (oldest anchor with below-median mass; fallback fifo) behaves like fifo ($\pm$1.5 points): at M=2 its stack is $\{\mathcal{A}_1, \mathcal{A}_T\}$ — the oldest anchor's mass is always above median and survives, everything in between dies at birth. Table 5 summarizes the resulting 2$\times$2: the primary key, not the secondary signal, determines whether a criterion works.

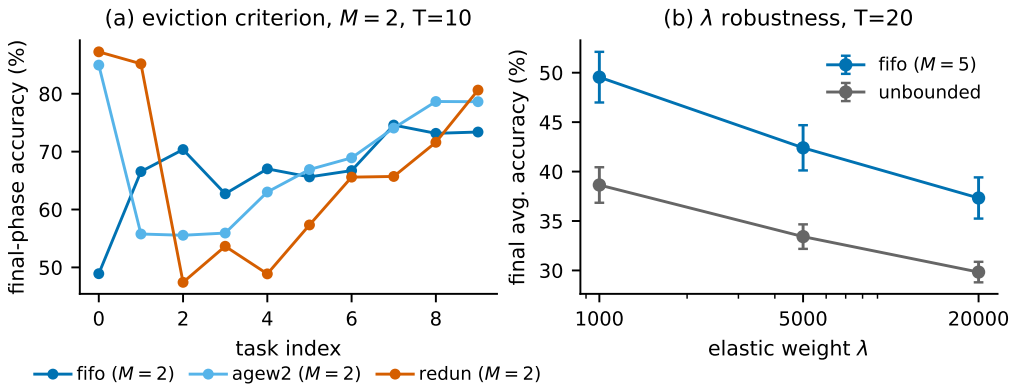


Figure 3: (a) Mass-keyed eviction (redun) grandfather-protects the first M tasks and erases the middle; age-first criteria (fifo, agew2) spread the cost. (b) The fifo advantage over unbounded holds at every λ.

5.6. Memory without behavior change: merge

Merging the most Fisher-similar pair removes exactly the cross term between the merged anchors. Penalty instrumentation over 40 runs ($T \in \{10, 20\}$, $M \in \{2, 5\}$, n=10; exactly $T-M$ merges per run) makes the change quantitative: the penalty's *value* drops one-sidedly — never increases — with per-run maximum relative drops of 43–45% at M=2 (worst case 64%) and 18–34% at M=5, and 77% of individual merges change that value. The change is *value-level only*: merging preserves the first-order quadratic regularizer (the gradient ∇_θ of the penalty is unchanged), shifting only the parameter-independent constant, which has zero effect on SGD. The removed mass is precisely the redundancy the merge detected; the accuracy trajectory nonetheless remains indistinguishable from unbounded at every M (Table 3) while storing $M \ll T$ anchors (-75% at T=20, M=5). Merge and fifo are complementary: merge buys storage at a measured, behaviorally invisible

Table 5: Criteria 2×2: primary key (rows) × secondary signal (columns). Only age-first criteria avoid the trap. T=10/T=20 BWT and accuracy, M=2 or M=5 as marked, n=3.

Primary key	Secondary	Outcome
mass	—	redun: grandfathering, BWT −30.2 (M5, T20)
mass	× age	agew ≡ redun (bit-identical)
age	—	fifo: sliding window, BWT −10.2 (M5, T20)
age	median mass	agew2 ≈ fifo (±1.5 pts)

penalty reduction but recovers no plasticity; fifo buys plasticity at a BWT price but no storage reduction.

5.7. Generality check: Fashion-MNIST

Fashion-MNIST, T=10, n=7 paired seeds (3–9): unbounded 35.6 ± 1.1; fifo M=5 38.4±1.6 (+2.8, $t(6)$=10.5); fifo M=2 48.6±2.1 (+13.0, $t(6)$=16.2). The budget effect replicates and amplifies as the budget shrinks.

5.8. Generalization to Split-CIFAR-100

Protocol.. Split-CIFAR-100, $T \in \{10, 20\}$: the 100 classes are split into T contiguous slices of $100/T$ classes each (10 classes/task at T=10, 5 classes/task at T=20; the code partitions a random permutation of the 100 classes into T blocks), ResNet-18 adapted to 32×32 inputs with GroupNorm in place of BatchNorm — running statistics are buffers that drift under later tasks and silently destroy protected tasks while the anchored weights themselves are held fixed, so they are excluded by design. Evaluation and training are task-incremental with a masked softmax head: each task computes its loss

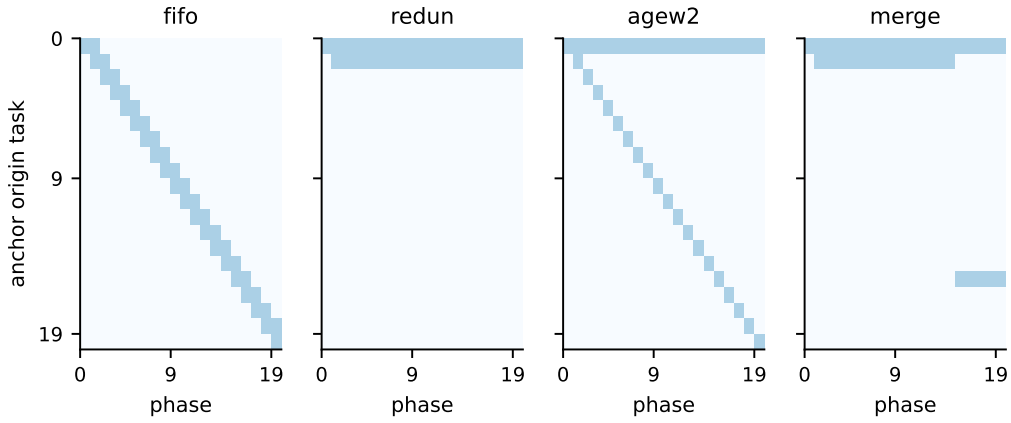


Figure 4: Anchor-stack composition over the task sequence (T=20, M=2, seed 0; identical for other seeds). Each column lists which anchors (by creation task) are alive at the end of each phase. fifo is a pure sliding window; redun freezes on the first two anchors (evicting every later anchor at birth); agew2 protects the oldest and the newest; merge partially grandfather-protects.

only over its own $100/T$ classes, so head rows of past tasks receive no gradient and are protected by construction; anchors therefore cover the backbone only, and forgetting can propagate exclusively through the shared trunk — exactly the pathway the anchor stack governs. Fisher diagonals are rescaled to unit global mean (one scalar over all parameters), making λ backbone-independent; gradients are clipped at norm 5. SGD, five epochs per task, λ=0.1 (chosen by an independent calibration sweep on unbounded DeLP only), n=10 seeds; eviction policies, origin instrumentation, and paired evaluation identical to the MNIST experiments. Unbounded DeLP never evicts, so its M is inert (final stack size T in every run); it is materialized once per T and serves as the reference for both budgets. All CIFAR comparisons are task-incremental (masked head); absolute accuracies are therefore not com-

Table 6: Split-CIFAR-100, λ=0.1: final accuracy, mean $\pm$ sd over n=10 seeds; parenthesized values are the paired difference Δacc against unbounded DeLP at the same seed (quoted in text with the paired $t(9)$ and the two-sided Wilcoxon signed-rank p-value).

	T=10		T=20	
Policy	M=2	M=5	M=2	M=5
unbounded	23.3 ± 2.6		29.8 ± 4.2	
fifo	24.8 ± 2.8 (+1.5)	23.5 ± 2.6 (+0.2)	$\mathbf{34.8 \pm 4.8}$ (+5.0)	32.2 ± 4.3 (+2.4)
random	24.7 ± 2.4 (+1.5)	22.4 ± 2.7 (−0.9)	33.4 ± 3.9 (+3.6)	32.2 ± 4.7 (+2.4)
redun	24.0 ± 2.5 (+0.7)	23.6 ± 2.0 (+0.3)	33.2 ± 4.7 (+3.4)	32.9 ± 3.3 (+3.1)
merge	22.3 ± 1.7 (−1.0)	21.8 ± 3.2 (−1.5)	30.8 ± 5.2 (+1.0)	29.3 ± 3.4 (−0.5)
agew	24.0 ± 3.8 (+0.7)	22.7 ± 3.4 (−0.6)	34.0 ± 4.3 (+4.2)	31.6 ± 3.7 (+1.8)
agew2	23.7 ± 2.6 (+0.4)	22.3 ± 3.3 (−1.0)	32.5 ± 3.7 (+2.7)	31.1 ± 3.4 (+1.3)

parable to the class-incremental Split-CIFAR-100 literature, and every claim in this section is a within-family, same-protocol paired comparison.

The budget effect strengthens with T on natural images.. At T=20 and the tightest budget (M=2), five of the six bounded policies outperform unbounded stacking (Table 6): fifo +5.0 ($t(9)$=7.1, Wilcoxon p=0.002), agew +4.2 (t=7.3, p=0.002), random +3.6 (t=9.2, p=0.002), redun +3.4 (t=4.0, p=0.006), agew2 +2.7 (t=2.7, p=0.037). At the looser budget (M=5) merge is the one policy that does not (−0.5), while the other five remain positive (+1.3 to +3.1); merge trades a small BWT reduction for storage and recovers no plasticity. At T=10 the effect is marginal, about +1.5 points at $t(9) \approx 2$, and no policy reaches significance under the Wilcoxon signed-rank test. The dose–response matches the mechanism: anchor drag accumulates with sequence length, so the benefit of bounding it should grow with T. As

in Sec. 5.1, the gain lands in the learning term: fifo M=2 raises ℓ from 31.4 to 36.8 while BWT moves only from -1.7 to -2.1 — the budget buys plasticity, the trade Eq. 1 predicts. The decomposition itself is exact on these runs: over all 260 CIFAR records, $|\bar{a}_T - (\ell + \frac{T-1}{T}\text{BWT})|$ never exceeds 5×10^{-3}, i.e. the rounding of two-decimal reporting.

5.9. A no-protection reference reorders the picture (Q4)

λ=0 (plain SGD, same protocol, n=10 paired seeds) reorders the picture and must be reported: 27.0±1.7 at T=10 and 36.9±3.4 at T=20 — above every DeLP arm, including fifo M=2, the strongest (+2.2, $t(9)$=2.19, p=0.057, Wilcoxon p=0.027 at T=10; +2.1, $t(9)$=1.46, p=0.18, Wilcoxon p=0.16 at T=20; the rank test, robust to the two large positive per-seed deltas, resolves the T=10 gap that the t-test only approaches). Against unbounded DeLP the margin is larger still and significant under both tests (+3.7, Wilcoxon p=0.006 at T=10; +7.1, Wilcoxon p=0.002 at T=20). The decomposition explains why: protection roughly halves the learning term (ℓ is 25.7 and 31.4 at T=10/20 for unbounded DeLP, vs. 46.1 and 64.3 at λ=0), while in this regime retention is nearly free to forgo — even the large BWT of the unprotected arm ($-21.3/-28.9$) costs less than the ℓ it gains. Under task-incremental evaluation with a fixed masked head, trunk features are shared across tasks and trunk drift is cheap; regularization-based protection at any λ is then a net tax on accuracy. The budget effect remains real within the protected family, but on this benchmark bounding is necessary, not sufficient: protection itself must be checked against a no-protection reference before the mechanism is credited.

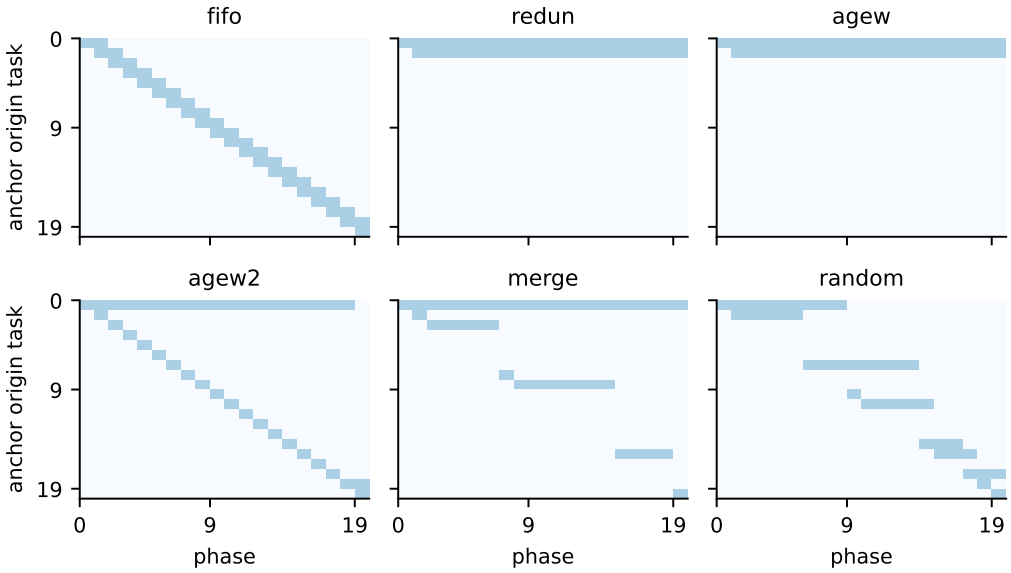


Figure 5: Anchor-stack composition on Split-CIFAR-100 (T=20, M=2, seed 0; all seeds agree on the final sets). Same rendering as Fig. 4. fifo is a sliding window; redun and agew freeze on the first anchors (grandfathering); agew2 pairs the oldest with the newest; merge partially grandfather-protects; random walks down.

Stack traces replicate bit-for-bit.. At T=20, M=2, all ten seeds agree: fifo's stack is the sliding window $\{\mathcal{A}_{T-1}, \mathcal{A}_T\}$; redun freezes on $\{\mathcal{A}_1, \mathcal{A}_2\}$ (grandfathering); agew inherits the same freeze (eviction-time mass swamps the age factor); agew2 holds $\{\mathcal{A}_1, \mathcal{A}_T\}$ in 6/10 seeds (the age-primary hybrid); merge grandfather-protects partially. The mechanism isolated on Permuted-MNIST (Fig. 4) is not an artifact of that domain (Fig. 5).

A boundary condition on the mass-keyed failure.. Unlike on MNIST, the *secondary* effect — *which* criterion evicts — is weak here: fifo − redun is +1.6 ($t(9)$=2.0) at T=20, M=2, and agew2 − redun is slightly negative. The reason is structural: under masked-CE training, forgetting propagates only

through the shared trunk, whose low layers are described similarly by every task's Fisher diagonal; even the stale anchors $\{\mathcal{A}_1, \mathcal{A}_2\}$ constrain features later tasks still use, so grandfathering them costs little. Mass-keyed eviction fails when anchors describe task-specific structure that goes stale; the primary effect of bounding (recovering plasticity) survives, while the relative cost of grandfathering scales with how much low-level structure tasks share — a testable prediction for domain-incremental sequences, where the gap should re-open.

6. Discussion

Why bounding works.. Eq. 1 says total retention is learning plus transfer. Unbounded stacks tax learning with stale constraints. A FIFO window keeps only the freshest M constraints, whose $\theta^{(k)}$ are near θ^{now} and whose Fisher mass is relevant to the current basin: the penalty constrains what actually needs constraining.

The mass signal is not useless — it is mis-keyed.. redun asks "which anchor constrains least *now*?"; the honest answer at eviction time is "the one just created", which is exactly the wrong anchor to drop. agew2 shows mass adds marginal, unstable value as a secondary signal. The information structure of the problem makes age the only reliable primary key: eviction-time mass confounds relevance with staleness.

Protection state as a knowledge base.. The perspective that unifies these findings is to treat the anchor stack as a knowledge base rather than an accumulator. A knowledge base is expected to state *what* it knows, *since when*, and

to support explicit maintenance operations — eviction, merge, audit — over named entries. An unbounded anchor stack violates all three expectations: it grows without bound, weights all entries equally regardless of age, and offers no interface for inspection. The bounded-stack family restores them: each anchor carries an origin tag, the survivor history records every maintenance event, the eviction policy is an explicit, swappable choice, and the log-linear law of Sec. 5.3 prices the knowledge retained per horizon unit. In this reading, the anchor accumulation problem is not an implementation bug in DeLP; it is what happens to *any* protection scheme whose knowledge base lacks a curation policy — and the same reading explains why the discounted posterior of online EWC, which answers the same curation question with an implicit decay, is interchangeable with the explicit window in accuracy but inferior in auditability.

Limitations.. (1) Task-incremental sequences (permuted MNIST, Fashion-MNIST, Split-CIFAR-100 with a masked head); domain-incremental settings, where tasks share nearly all low-level structure, may shrink the age-over-mass gap further (Sec. 5.8). (2) $T \leq 20$; the ℓ-recovery trend grows with T but the BWT cost does too. (3) All MNIST arms share an uncalibrated hidden gain, as in the audit; the CIFAR study uses its own protocol with λ calibrated on unbounded DeLP only; conclusions are within-family comparisons. (4) Merge's penalty change is measured by instrumentation (Sec. 5.6), not bounded formally; a closed-form error bound is future work. (5) On Split-CIFAR-100 the no-protection reference outperforms every protection arm (Sec. 5.8): task-incremental evaluation makes forgetting cheap, so the anchor pathway is a net tax there and the budget result is internal to pro-

tected arms. (6) The three benchmarks are the standard continual-learning benchmarks of the regularizer literature, not large-scale vision models: we do not evaluate ImageNet-scale class-incremental streams, so the magnitude of the anchor-accumulation cost in that regime is unmeasured. We regard the diagnosis, not the repair, as the transferable claim here — the accumulation law of Sec. 5.3 follows from the regularizer's form rather than from the task distribution, and should therefore scale; establishing the corresponding magnitudes at scale is the next step, and the instrumentation used here (anchor-stack survivor histories, per-run penalty traces) is designed to be portable to that setting.

7. Conclusion

Bounding the anchor stack converts the diagnosed anchor accumulation problem into a control knob: FIFO eviction recovers the learning term at a declared BWT cost, the budget's effect is structurally different from λ's, age is the only sound eviction signal, and merge decouples memory from behavior. The effects replicate across three benchmarks — permuted and class-incremental, MLP- and ResNet-based — with the age-over-mass advantage bounded by a stated boundary condition on task relatedness, and the protection pathway itself bounded by a no-protection reference on Split-CIFAR-100. Single-anchor schemes cannot accumulate the problem by construction, and a discounted posterior — itself a soft bound, its mass converging to a horizon of $\approx 1/(1-\gamma)$ tasks — recovers plasticity through the same law: across soft and hard bounds the learning term falls log-linearly with the protected-mass horizon at the calibrated λ (monotone at every grid value), and a tuned dis-

count overtakes the window on accuracy at matched forgetting. Bounding accumulated anchor mass, not the eviction mechanism, is the active ingredient; the explicit window is the variant whose protection set is auditable and whose policy is a choice. Protection schemes for continual learning should ship with bounded — or at least mergeable — anchor state whenever protection is warranted at all; the Split-CIFAR-100 reference (Sec. 5.8) shows that the warrant itself is regime-dependent and must be checked first.

CRediT authorship contribution statement

Ranxi Zhou: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Visualization.

Declaration of competing interest

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Data availability

All code, experiment scripts, and the complete JSON records of every run (including per-phase accuracy matrices and anchor-stack sur-

vivor histories) are publicly available at `https://github.com/zhouranxi/delp-anchor-budget`. This manuscript and its full reproducibility package — including the audit scripts that re-derive every quoted number from those records — are archived on Zenodo (doi:10.5281/zenodo.23668522). The precursor audit study is archived at doi:10.5281/zenodo.23572226 and the companion segment-specialization audit at doi:10.5281/zenodo.23637972; the underlying dataset and code release is doi:10.5281/zenodo.23639561.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used AI-assisted language-editing tools in order to improve the grammar, clarity, and consistency of the manuscript text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article. The use of AI tools did not extend to any scientific content: all experimental design, analysis, figures, and results were produced and verified by the author.

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